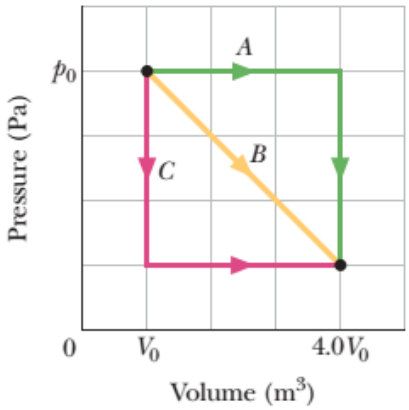
Physics 2 Assignment Problems for Mid-term
Spring-2019-20
Date of Submission: 25/2/2020 for Section-E and 26/2/2020 for Section D, H

1. What mass of steam at 100^0 C must be mixed with 150 g of ice at its melting point, in a thermally insulated container, to produce liquid water at 50^0 C?
2. A small electric immersion heater is used to heat 300 g of water for a cup of instant coffee. The heater is labeled "440 watts" (it converts electrical energy to thermal energy at this rate). Calculate the time required to bring all this water from 13.0 ^{0}C to 100 ^{0}C, ignoring any heat losses.
3. A certain substance has a mass per mole of 23.2 g/mol. When 300.25 J is added as heat to a 30.0 g sample, the sample's temperature rises from 15.0 ^{0}C to 55.0 ^{0}C. What are the (a) specific heat and (b) molar specific heat of this substance? (c) How many moles are in the sample?
4. Calculate the minimum amount of energy, in joules, required to completely melt 130 g of Aluminum initially at 15.0 ^{0}C. Specific heat capacity of Aluminum is 921.096 $jkg^{-1}k^{-1}$ and melting point of Aluminum is 660.3^0C.
5. In Fig., a gas sample expands from V_0 to $4.0V_0$ while its pressure decreases from p_0 to $p_0/4.0$. If $V_0 = 0.6$ m^3 and $p_0 = 30.6$ Pa, how much work is done by the gas if its pressure changes with volume via (a) path A, (b) path B, and (c) path C?



6. How much heat must be absorbed by ice of mass m =720 g at -10^0C to take it to the liquid state at 15^0C
7. A copper slug whose mass m_c is 75 g is heated in a laboratory oven to a temperature T of 312°C. The slug is then dropped into a glass beaker containing a mass m_w =220 g of water. The heat capacity C_b of the beaker is 45 cal/K. The initial temperature T_i of the water and the beaker is 12°C.Assuming that the slug, beaker, and water are an isolated system and the water does not vaporize, find the final temperature T_f of the system at thermal equilibrium.
8. Let 1.00 kg of liquid water at 100^0C be converted to steam at 100^0C by boiling at standard atmospheric pressure (which is 1.00 atm or 1.01 x 10^5 Pa) in the arrangement of

The volume of that water changes from an initial value of 1.00 x10^3 m^3 as a liquid to 1.671 m^3 as steam. How much work is done by the system during this process?

9. A cylinder contains 12 L of oxygen at 20^0 C and 15 atm. The temperature is raised to 35^0C, and the volume is reduced to 8.5 L. What is the final pressure of the gas in atmospheres? Assume that the gas is ideal.

10. A bubble of 5.00 mol of helium is submerged at a certain depth in liquid water when the water (and thus the helium) undergoes a temperature increase ΔT of 20.0^0C at constant pressure. As a result, the bubble expands. The helium is monatomic and ideal.

11. Initially an ideal diatomic gas has pressure *pi* =2.00 x10^5 Pa and volume V_i = 4.00 x 10^6 m^3. How much work *W* does it do, and what is the change *E*int in its internal energy if it expands adiabatically to volume V_f =8.00 x 10^6 m^3? Throughout the process, the molecules have rotation but not oscillation.

12. Initially, 1 mol of oxygen (assumed to be an ideal gas) has temperature 310 K and volume 12 L. We will allow it to expand to volume 19 L. What would be the final temperature if the gas expands adiabatically? Oxygen (O_2) is diatomic and here has rotation but not oscillation.

13. Oxygen gas having a volume of 1000 cm^3 at 40.0 C and 1.01 x 10^5 Pa expands until its volume is 1500 cm^3 and its pressure is 1.06 x 10^5 Pa. Find (a) the number of moles of oxygen present and (b) the final temperature of the sample.

14. The best laboratory vacuum has a pressure of about 1.00 x 10^{18} atm, or 1.01 x10^{13} Pa. How many gas molecules are there per cubic centimeter in such a vacuum at 293 K?

15. The temperature of 2.40 mol of an ideal monatomic gas is raised 18.60 K at constant volume. What are (a) the work *W* done by the gas, (b) the energy transferred as heat *Q*, (c) the change ΔE_{int} in the internal energy of the gas, and (d) the change ΔK in the average kinetic energy per atom?

16. When 21.77 J was added as heat to a particular ideal gas, the volume of the gas changed from 30.0 cm^3 to 190 cm^3 while the pressure remained at 1.23 atm. (a) By how much did the internal energy of the gas change? If the quantity of gas present was 2.23x10^{-3} mol, find (b) C_p and (c) C_V.

17. We know that for an adiabatic process pV^γ = a constant. Evaluate “a constant” for an adiabatic process involving exactly 2.0 mol of an ideal gas passing through the state having exactly *p* = 1.9 atm and *T* = 329 K. Assume a diatomic gas whose molecules rotate but do not oscillate.

18. Imagine a Carnot engine that operates between the temperatures T_H = 750 K and T_L =350 K. The engine performs 1600 J of work each cycle, which takes 0.29 s.

19. An inventor claims to have constructed an engine that has an efficiency of 75% when operated between the boiling and freezing points of water. Is this possible?

20. A Carnot engine operates between 235 ^{0}C and 135 ^{0}C, absorbing 4.30x10^4 J per cycle at the higher temperature. (a) What is the efficiency of the engine? (b) How much work per cycle is this engine capable of performing?